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FlexDelivery – Integrated Logistics in the Offshore Sector

Kris Vedat, P&O Maritime Logistics Limited, Dubai, United Arab Emirates

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Abstract

This case study aims to showcase how integrated logistics in the offshore sector streamline the supply chain process, reduce costs, and improves efficiency. The scope of integrated logistics includes planning, transportation, warehousing, inventory management, and information management, focusing on collaboration and transparency between all stakeholders in the offshore supply chain.

The process of integrated logistics in the offshore sector begins with the cargo booking. A detailed logistics plan and schedule are then developed, outlining the supply chain network, transportation modes, and inventory management strategies. The process is managed by an AI-based platform that automatically creates short and long-term schedules using various cargo and telemetric data. During the execution phase, real-time tracking and monitoring of the supply chain process are crucial to managing disruptions. Continuous improvement is key to optimising the integrated logistics process with a machine learning element to the logistics tool, resulting in increased efficiency, reduced costs, and improved safety and reliability.

Implementing integrated logistics in the offshore sector has yielded several positive results. Firstly, it has improved efficiency in the supply chain process, reducing the time and cost required to move goods and equipment from the point of origin to the point of consumption. Delivery time has been reduced by 23%, achieved by using an AI planning system, real-time tracking, and optimised transportation modes.

Secondly, integrated logistics has helped to maintain high levels of safety by reducing the number of entries into the 500M zone by consolidating cargo and increasing back deck utilisation. Standardised procedures for logistics operations have been established, minimising the risk of errors and improving overall safety.

Thirdly, the implementation of integrated logistics has led to increased collaboration and communication between stakeholders involved in offshore operations, resulting in improved decision-making and reduced delays, as well as better transparency between all elements of the supply chain.

Real-time tracking and monitoring of the supply chain process have been crucial for effectively managing disruptions and addressing issues, which is made possible by automating the process using AI, which is more efficient than manual processes.

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The use of integrated logistics in the offshore sector has resulted in an overall cost reduction of 23% on the shipment of goods and a reduction of CO2 emissions by 32%, enabling effective management of the movement of goods and equipment while promoting sustainability.

This approach to integrated offshore logistics will enable effective management of the movement of goods and equipment from the point of origin to the point of consumption and reduce costs for the oil and gas sector while ensuring compliance with regulatory requirements.

Introduction

As globalisation continues to grow, the importance of efficient logistics management has become increasingly vital. A company's ability to deliver goods and services in a timely and efficient manner is now a critical aspect of its competitive advantage. This has necessitated innovation and the implementation of technology in all sectors, including the offshore sector within the oil and gas industry.

The offshore sector within the oil and gas industry presents unique challenges. Often operating in harsh and remote environments, the sector must manage complex supply chains, including the transportation of equipment and materials from various points of origin to the final point of consumption. Any inefficiency or breakdown in the supply chain can lead to significant delays and cost overruns.

The goal was to re-imagine and reinvent the role of the supply chain, moving away from archaic operating models towards a more modern, efficient, and environmentally friendly approach. Central to this vision was the introduction of resource sharing and optimised cargo planning. The concept of shared vessels and facilities and paying only for space used would bring about economies of scale, reduce waste, and eliminate duplication of resources.

The solution offers a novel way to solve these challenges by providing an integrated approach to logistics in the offshore sector. As a comprehensive system, it merges various elements such as planning, transportation, warehousing, inventory management, and information management. The system aims to streamline the supply chain process, provide a more flexible logistics method to the oil and gas industry, reduce costs, and improve efficiency through a strategic and data-driven approach.

At the core of the solution lies artificial intelligence (AI) and machine learning (ML) technologies. These technologies are instrumental in managing the vast array of data involved in the supply chain process, from cargo booking to final delivery and are not new to the industry. AI algorithms are employed to create efficient schedules, and ML techniques have been used to learn from historical data and continuously improve processes over a variety of domains [1 – 6].

The system's strength lies in its ability to seamlessly integrate the various elements of logistics management. The planning phase uses AI to create comprehensive logistics plans. Transportation is optimised using real-time tracking systems, and inventory management is improved through data-driven strategies. Warehousing is streamlined with the help of efficient layout plans, and information management ensures transparency across the entire supply chain process. The focus is not only on individual elements but also on their interconnectivity to ensure a smooth and efficient process.

Methods

The operational process of the solution begins with booking cargo. This is an essential first step that involves gathering information about the goods to be transported, including weight, size, quantity, and other specific handling requirements. This data serves as the foundation for the subsequent steps of the logistics process.

Once the cargo has been booked, an AI-based platform uses the cargo data, along with telemetric data, to generate an in-depth logistics plan and schedule. This AI-driven system is capable of considering numerous variables simultaneously and can generate both short and long-term schedules. It optimises various elements, including route selection, transportation modes, and inventory management strategies.

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During the execution phase, real-time tracking and monitoring become essential. This continuous tracking allows the platform to account for any potential disruptions, such as weather changes or technical issues, and adjust the logistics plan accordingly. It also enables transparency in the process, providing stakeholders with the ability to monitor the progress of their goods throughout the supply chain.

The system employs machine learning to continually improve the logistics process. By learning from past data, the system can predict and prepare for potential issues, resulting in increased efficiency, reduced costs, and improved safety and reliability. This continuous learning and adapting capability make the system dynamic and increasingly robust over time.

The solution addresses numerous challenges present in the offshore sector. Traditional operating models often involve duplication of manpower, facilities, and equipment, leading to inefficiency and waste. By consolidating cargo and optimising the use of vessels, The system reduces unnecessary duplication and maximises utilisation. Furthermore, it digitises operations, replacing paper-based procedures with a streamlined, technology-driven approach. This approach enables more efficient data management, transparency, and collaborative decision-making.

Fig. 1 showcases the information workflow of the system, illustrating the strategic integration of various elements such as planning, transportation, warehousing, inventory management, and information management. The figure also highlights how AI and Machine Learning are incorporated within this workflow to optimise operations. The flow of information is depicted from the booking of cargo to real-time tracking, with an emphasis on the iterative process of continuous improvement driven by AI and Machine Learning. The figure highlights the importance of data-driven decisions in modern logistics, demonstrating the system's capacity to provide timely, accurate information for effective management of the supply chain.

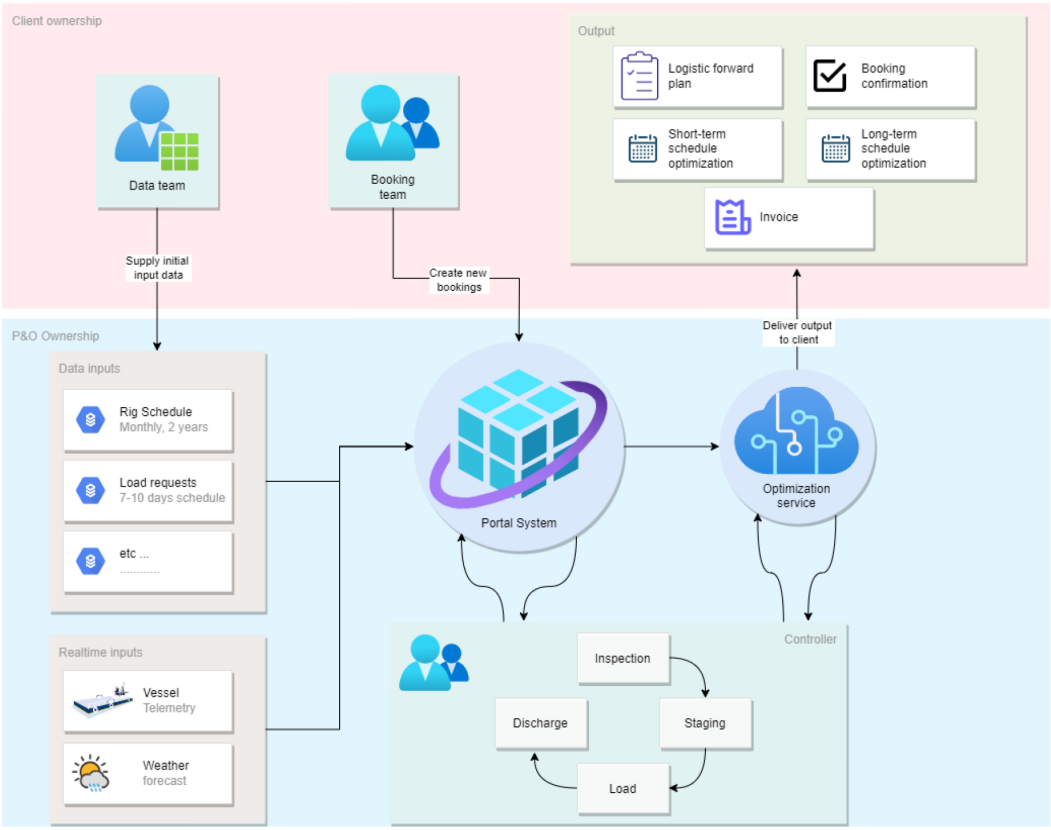

Figure 1—Information Workflow

Fig. 2 outlines the operational process of the solution, providing a step-by-step visualisation of the integrated logistics process from cargo booking to the actual delivery. This figure underscores the critical

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role of AI in automating and enhancing various stages of the logistics process, from planning and scheduling to real-time tracking and disruption management. It demonstrates the system's ability to streamline operations, ensure efficient resource utilisation, and facilitate seamless coordination between various stakeholders. Furthermore, it exemplifies how the system adopts a proactive approach to address potential disruptions and enhance safety measures, contributing to the overall efficiency and reliability of the supply chain in the offshore sector.

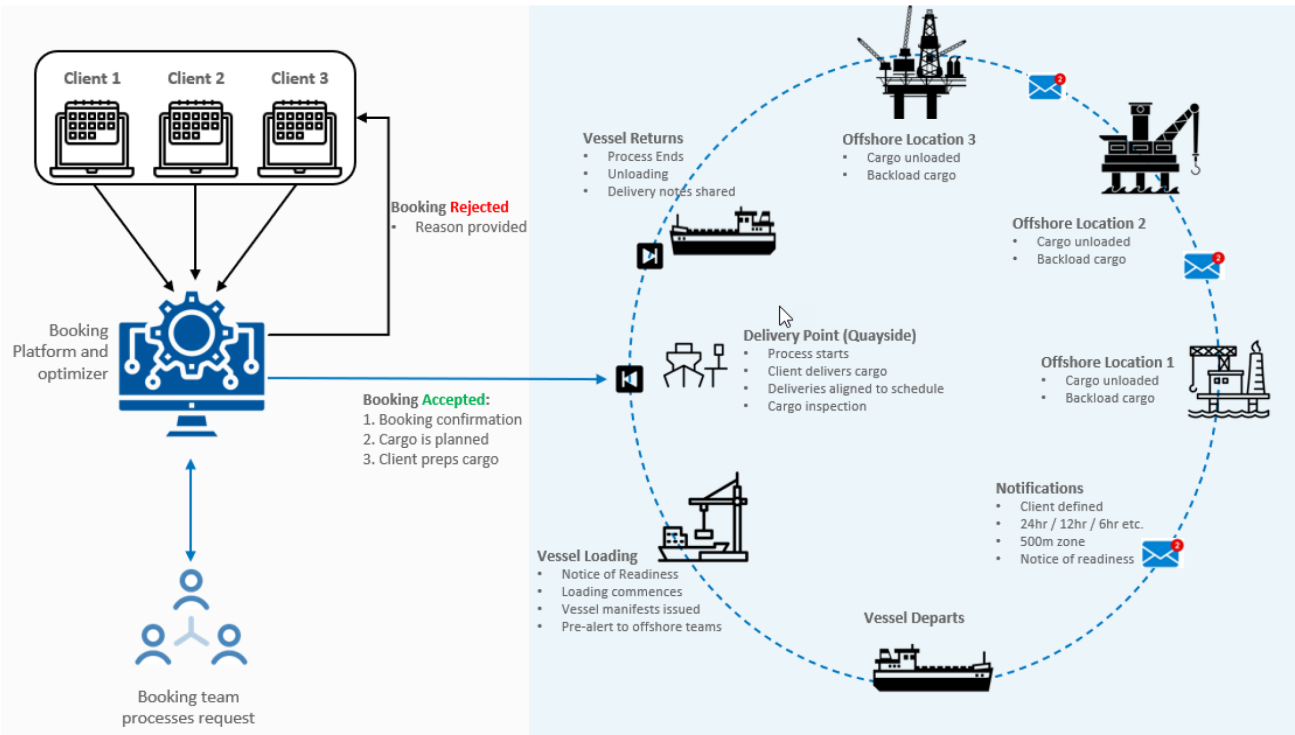

Figure 2—Operational Process

Addressing Challenges

The offshore sector has historically been plagued with numerous challenges that have hampered efficiency and increased costs. The system has been designed with the aim of overcoming these challenges through an innovative approach to integrated logistics.

1. **Overcoming Archaic Operating Models**: The first major challenge is the persistence of outdated, inefficient operating models. The systems approach, which employs AI and machine learning technologies, represents a paradigm shift from these traditional models. Automating processes and leveraging data for decision-making brings a modern, technology-driven approach that vastly improves efficiency and reduces costs.
2. **Reducing Duplication of Resources**: The system also addresses the problem of duplication of manpower, facilities, and equipment in the offshore sector. Its per-square-meter cost model encourages companies to maximise vessel utilisation, as they only pay for the space they use. This eliminates the need for redundant resources and reduces unnecessary costs. The use of shared vessels and fixed sailing schedules further promotes efficient use of resources and cost savings.
3. **Transitioning from Paper-Based to Digital Operations**: A key aspect of the solutions approach is the transition from paper-based operations to digital systems. This not only speeds up processes and reduces the potential for errors but also enables real-time tracking and monitoring of the supply

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chain process. It addresses the offshore sector's traditional lack of supply chain technology and allows for greater transparency and improved decision-making.

4. **Optimising Underutilised Systems**: Many systems in the offshore sector have historically been underutilised, inefficient, and wasteful. The systems AI-based platform optimises these systems by intelligently planning and scheduling logistics operations. It also promotes route optimisation, resulting in significant fuel savings and reduced CO2 emissions.
5. **Promoting Standardisation**: Lastly, the system encourages the standardisation of logistics operations, which helps to minimise the risk of errors and improve overall safety. Standardised procedures also simplify processes, making it easier to train staff and maintain high levels of operational consistency.

In summary, the solution innovative approach addresses the various challenges facing the offshore sector, providing a roadmap for other industries dealing with similar issues. It offers a holistic solution that streamlines operations, optimises resources, and harnesses technology to deliver cost-effective, efficient, and environmentally friendly logistics.

Results

The system has proven to increase efficiency in the supply chain process significantly. A notable achievement has been a 23% reduction in delivery time, made possible by the AI planning system, real-time tracking, and optimised transportation modes. This reduction significantly lowers the time required to move goods and equipment from the point of origin to the point of consumption.

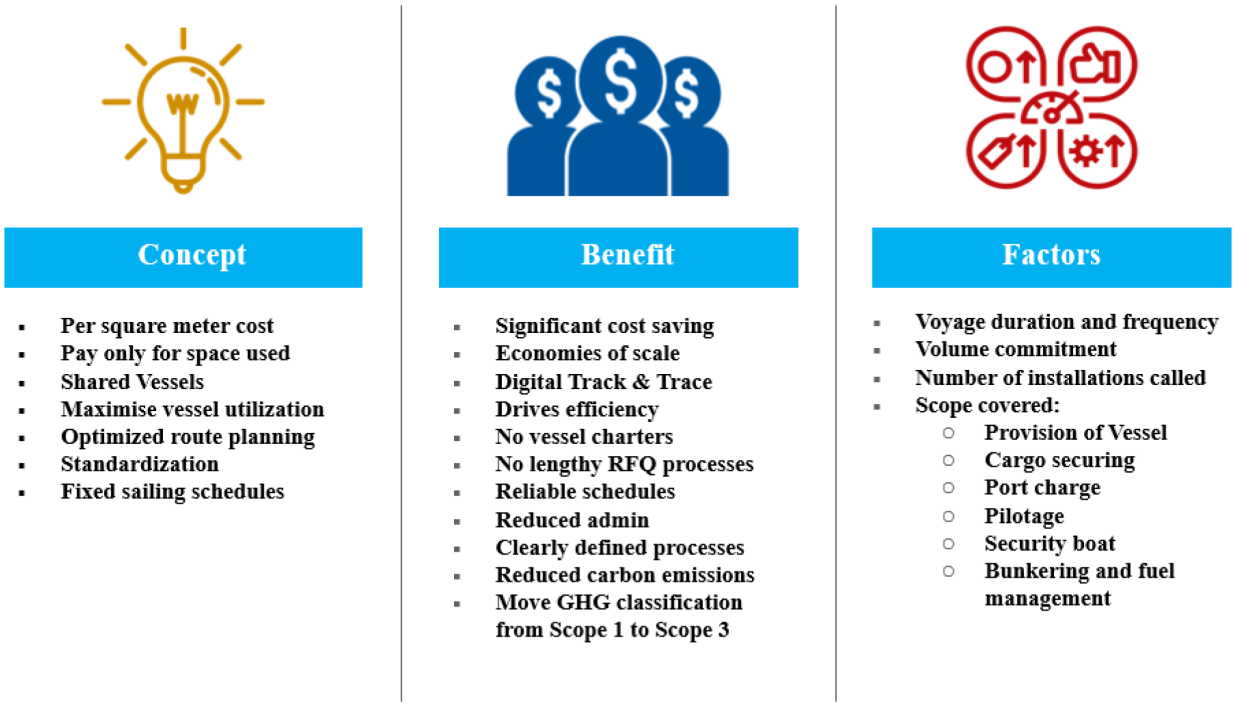

Figure 3—The Concept, Benefit and Factors Overview

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Improved collaboration and communication among stakeholders are other significant benefits of implementing the system. By providing real-time tracking and transparency in operations, decision-making is improved, and delays are reduced. This enhanced collaboration contributes to more efficient and streamlined operations.

In traditional offshore logistics operations, communication and collaboration between stakeholders are often challenging due to a lack of real-time visibility and transparency into operations. These challenges can lead to inefficiencies, such as delays in decision-making, misalignment of operations, and a lack of coordination, ultimately resulting in increased costs.

The AI-based platform, addresses these issues by providing a single, unified platform where all the necessary information is readily available and easily accessible to all relevant parties. This includes data on cargo, transportation schedules, vessel locations, and real-time updates on the execution phase. With this level of transparency, all stakeholders have a clear and accurate picture of the ongoing operations, leading to improved decision-making. Moreover, the system's real-time tracking and monitoring capabilities facilitate seamless communication between different stakeholders. For instance, if there's a change in the weather that could potentially disrupt transportation, all parties are immediately informed. This level of communication allows for quick response to any changes, reducing potential delays and enhancing operational efficiency.

Furthermore, the system promotes collaboration by providing a shared platform where stakeholders can work together towards a common goal. Companies can easily coordinate their operations, align their schedules, and collaborate on decision-making through this platform. For example, suppose two companies have cargo that needs to be transported to the same location. In that case, they can coordinate their schedules to consolidate their cargo, thereby maximising vessel utilisation and reducing costs.

The solution also fosters long-term partnerships among stakeholders. By providing a reliable and efficient logistics solution, it encourages companies to work together more closely and form strategic partnerships. These partnerships can lead to further efficiency improvements, cost savings, and better alignment of operations.

Case Study

This case study examines the significant cost, environmental, and efficiency benefits achieved by two offshore installation clients through the implementation of the service. Historically operating with high expenses, considerable fuel consumption, and extensive CO2 emissions, the implementation of the system transformed the clients' operations. The study, Table 1 -3 and the impact section below present tangible results, highlighting a remarkable decrease in costs, a substantial reduction in fuel usage and CO2 emissions, and significant efficiency improvements in terms of distance travelled and time saved. The findings illustrate the substantial impact that logistics optimisation strategies like this system can have on the offshore industry.

Table 1—Case Study – Traditional Approach

Run	PoL	To	PoD	Dist. (nm)	Duration (Hrs)
1	Load Port	Location 1	Discharge Port	132	17
2	Load Port	Location 2	Discharge Port	183	23
3	Load Port	Location 3	Discharge Port	211	26
4	Load Port	Location 4	Discharge Port	217	27
5	Load Port	Location 5	Discharge Port	327	41
				1,070	134

Client Name	Sqm per Voyage	2 day Independent Vessel run	No. of Locations Serviced	Total Cost	Total Sqm	Cost Per Sqm
Client 1	200	$ 130,000.00	3.00	$ 390,000.00	600.00	$ 650.00
Client 2	200	$ 130,000.00	2.00	$ 260,000.00	400.00	$ 650.00
			Total	$ 650,000.00		

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Table 2—Case Study – The Approach

Legs	From	To	Dist. (nm)	Duration (Hrs)
1	Load Port	Location 1	64	8
2	Location 1	Location 2	27	3
3	Location 2	Location 3	43	5
4	Location 3	Location 4	11	1
5	Location 4	Location 5	197	25
6	Location 5	Discharge Port	185	23
			527	66

Client Name	Sqm per Voyage	No. of Locations Serviced	Total Sqm	Cost per Sqm FlexDELIVERY	7-Day Run FlexDELIVERY	Saving per Voyage
Client 1	200	3.00	600.00	$ 500.00	$ 300,000.00	$ 90,000.00
Client 2	200	2.00	400.00	$ 500.00	$ 200,000.00	$ 60,000.00
			1,000.00		$ 500,000.00	$150,000.00

Table 3—Case Study – The Impact

Run	PoL	To	PoD	Dist. (NM)	Voyage in days	7 Fuel cons. in MT per run	3.25 CO2 Emissions in MT per run	60.00 CO2 offset cost in MT per run
1	Load Port	Location 1	Discharge Port	132	2	14	46	$ 2,730.00
2	Load Port	Location 2	Discharge Port	183	2	14	46	$ 2,730.00
3	Load Port	Location 3	Discharge Port	211	2	14	46	$ 2,730.00
4	Load Port	Location 4	Discharge Port	217	2	14	46	$ 2,730.00
5	Load Port	Location 5	Discharge Port	327	3.0	21	68	$ 4,095.00
			Totals for 5 Voyages	1,070	11.0	77.0	250.3	$ 15,015.00

Legs	From	To	Dist. (NM)	Leg in days	7 Fuel cons. in MT per Leg	3.25 CO2 Emissions in MT per leg	60 CO2 offset cost in MT per Leg
1	Load Port	Location 1	64	1	7	22.75	$ 1,365.00
2	Location 1	Location 2	27	1	7	22.75	$ 1,365.00
3	Location 2	Location 3	43	1	7	22.75	$ 1,365.00
4	Location 3	Location 4	11	1	7	22.75	$ 1,365.00
5	Location 4	Location 5	197	1	7	22.75	$ 1,365.00
6	Location 5	Discharge Port	185	2	14	45.5	$ 2,730.00
Totals for 1 Voyage covering 6 Legs			527	7	49	159	$ 9,555.00

Impact

Cost-Reduction

In addition, the implementation has resulted in significant cost savings in offshore logistics operations. The use of AI and machine learning, coupled with the implementation of integrated logistics, has resulted in an optimised supply chain. This optimisation is evident in several ways.

Firstly, it has enabled a 23% reduction in the cost of goods shipment. This significant decrease has been achieved through various means. For instance, real-time tracking and monitoring, coupled with AI-

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generated short and long-term schedules, have optimised the transportation process, reducing the delivery time and, consequently, the cost associated with it.

Secondly, by reducing the number of entries into the 500M zone through cargo consolidation and increased back deck utilisation, there's a decrease in operating costs. Minimised risk of errors from standardisation of procedures also reduces the potential for costly mistakes.

Thirdly, better collaboration among stakeholders minimises delays, further reducing costs associated with prolonged operations or idle times. The ability to coordinate and plan together can lead to shared resources and consolidated efforts, leading to cost efficiency.

Overall, the system provides an efficient way to reduce the cost per square meter from $650 to $500, offering substantial savings. In one example, the original cost of $650,000 was optimised to $500,000, a cost-saving of $150,000, or 23%.

Sustainability

In addition to cost reduction, the solution has made significant strides in enhancing the sustainability of offshore operations. The digitalised integrated logistics platform has been instrumental in reducing the environmental impact of operations in several ways.

Firstly, it has resulted in a 32% reduction in CO2 emissions. By optimising routes, consolidating cargo, and reducing the number of entries into the 500M zone, minimises fuel consumption, which in turn reduces CO2 emissions.

Secondly, the use of real-time tracking and monitoring allows for more precise operations, reducing unnecessary manoeuvring and minimising waste. The AI-based system's ability to predict and plan the most efficient routes further ensures minimal fuel consumption and, therefore, minimal emissions. By optimising transportation schedules and routes, consolidating cargo, and maximising back deck utilisation, the AI-based platform can significantly reduce fuel consumption. The data provided in the study reveals that fuel consumption decreased from 77 MT to 49 MT, representing a remarkable drop. The case study also reports a reduction of CO2 emissions from 250 MT to 159 MT, amounting to a total reduction of 91 MT or 36%. This reduction not only signifies cost savings but also aligns with global initiatives to reduce greenhouse gas emissions, highlighting the sustainability advantage. For every 1,000 square meters shipped, the reduction in CO2 emissions translates into savings of $5,460. This is significant, considering the scale at which offshore operations take place, and these savings add up to substantial amounts over time.

Moreover, by reducing the delivery time by 23%, less energy is expended per operation, which also contributes to the overall sustainability of the process. This efficiency has led to a significant reduction in distance travelled, from 1,070 nautical miles to 527 nautical miles, a reduction of 51%, saving 68 hours of sailing time.

Lastly, by providing a platform for better collaboration among stakeholders, The solution allows for shared resources and synchronised operations, reducing the redundancy and waste often seen in isolated operations.

Additionally, the sustainability benefits of the systems-integrated logistics approach have positive ripple effects on other areas of the offshore industry. Companies can better meet regulatory requirements and improve their environmental reputation by reducing environmental impact. This, in turn, can lead to increased trust and confidence from stakeholders, potentially opening up new business opportunities.

In conclusion, the adoption of the systems-integrated logistics approach leads to a more sustainable offshore operations environment. Not only does it reduce the sector's carbon footprint significantly, but it also ensures that resources are utilised efficiently, contributing to long-term sustainability in the sector.

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Procurement Behavior Change

Procurement is a crucial aspect of any logistics system, with direct impacts on cost, efficiency, and overall performance. The system influences procurement behaviour in several ways, mainly by promoting data-led decision-making and optimising material flows and operational programs.

Traditionally, procurement decisions in offshore logistics have been guided by experience, intuition, and, often, a degree of guesswork. However, the solutions AI and machine learning capabilities allow procurement teams to leverage data to make more accurate, informed decisions. For example, procurement teams can use data from the AI-based platform to identify the most efficient routes, the most cost-effective transportation modes, and the optimal times for shipments.

The system's predictive capabilities can also forecast future demand for various items, enabling procurement teams to adjust their ordering practices accordingly. This can prevent over-ordering and under-ordering, both of which can lead to increased costs and inefficiencies.

Furthermore, the systems focus on cargo planning and considers the relationship between material flows and the operational programs they support. This means procurement teams are better equipped to understand how their decisions will impact the entire supply chain. They can plan purchases in such a way that materials arrive just when they are needed, reducing the need for warehousing and reducing the risk of delays in the operational schedule.

By aligning procurement activities with the overall logistics strategy, the system enables companies to maximise the value of their procurement spend. This not only results in cost savings but also leads to improvements in operational efficiency and overall supply chain performance.

Discussion on the Systems Approach to Cargo Planning

Cargo planning is a critical aspect of supply chain management, particularly in the offshore sector, where the movement of goods and equipment often involves multiple modes of transportation and complex schedules. The system's innovative approach to cargo planning is a significant factor in its success and a major driver of its efficiency and cost savings.

One key feature of the solutions approach to cargo planning is its utilisation of an AI-based platform. This platform uses cargo and telemetric data to create both short and long-term schedules for the movement of goods. This sophisticated planning process considers various factors such as cargo size, weight, type, destination, and the most efficient transportation mode.

Through its AI and machine learning capabilities, The systems platform can optimise cargo consolidation, enabling multiple shipments to be combined into a single transportation mode. This approach maximises vessel utilisation and reduces the number of entries into the 500M zone, contributing to both operational efficiency and safety improvements.

The systems cargo planning approach also considers the relationship between material flows and the operational programs they support. This perspective ensures that the timing of cargo deliveries aligns with operational schedules, minimising downtime and maximising productivity. It also helps to reduce warehousing needs, as materials can be delivered just when they're needed.

Furthermore, the solutions AI-based platform offers real-time tracking and monitoring of the cargo. This allows for instant visibility into the status and location of the goods, enabling quick reactions to any disruptions and ensuring timely delivery. It also provides transparency to all stakeholders, leading to improved decision-making and collaboration.

Conclusion

As evidenced by this study, the system exhibits immense potential in transforming integrated logistics within the offshore sector. Its strategic use of artificial intelligence (AI) and digital solutions has redefined the parameters of efficiency, safety, and cost-effectiveness in offshore logistics operations.

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AI and digital solutions serve as the backbone of the platform, effectively addressing the industry's perennial challenges, such as operational inefficiencies, safety risks, and high costs. By leveraging these technologies, the systems infuses data-driven insights and automation into logistics processes, setting the stage for continuous improvement and optimisation.

The study further reveals that the platform's influence extends beyond enhancing efficiency. It has revolutionised safety and transparency within offshore logistics, primarily through its real-time tracking and monitoring features. These capabilities minimise risks and provide actionable data, enabling stakeholders to make informed decisions that advance their operational objectives.

Moreover, this strikes a balance between economic viability and environmental stewardship. Its integration of AI and machine learning has unlocked significant cost savings. Concurrently, the system's optimisation of resource planning and utilisation has dramatically reduced CO2 emissions, underscoring its commitment to sustainability.

The implications of the success of the system are profound for the oil and gas industry. With sustainability and cost-effectiveness as key imperatives in today's fast-paced world, the platform offers a new way for the industry to rethink its supply chain operations. The system sets a benchmark for future offshore sector operations by aligning with these imperatives.

In conclusion, the systems approach manifests the transformative potential of integrated logistics in the offshore sector. Through the effective use of AI and digital solutions, it tackles the industry's longstanding challenges head-on, catalysing improvements in efficiency, safety, and transparency. The system's delivery of substantial cost savings, coupled with its significant environmental benefits, underscores its value proposition. As such, the solution provides a compelling blueprint for re-imagining supply chain operations in the oil and gas industry, ensuring that they are both sustainable and cost-effective in an ever-evolving world.

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